

Advanced Migration Features

LUKS Encryption, LVM, Batch Orchestration, AI Diagnostics, Daemon Mode & More

Enterprise-grade features for complex migrations: LUKS2 + TPM encryption, container-isolated LVM handling, batch migration with checkpoint/resume, AI-powered diagnostics and remediation, daemon mode with file system watching, structured observability, and recovery with rollback.

LUKS2 + TPM | LVM Isolation | Batch + Resume | AI Engine | Daemon Mode | 17 Feature Flags

LUKS2 Encryption & Storage

Migrate VMs to encrypted disks with automatic TPM enrollment — zero-touch encryption for confidential compute.

LUKS2 + TPM Pipeline

- End-to-end: VMDK → LUKS2 container → filesystem migration
- AES-XTS-plain64 encryption, SHA256 hashing
- Argon2id PBKDF (memory-hard key derivation)
- Key management: keyfiles or TPM auto-unlock
- TPM2 enrollment for seamless boot
- initramfs rebuild with dm_crypt + xts modules
- Targets: OpenShift confidential VMs, Talos, Flatcar

Configuration

```
# Enable LUKS encryption
luks_enable: true
luks_passphrase: "encryption-key"
# Or use environment variable:
luks_passphrase_env: LUKS_KEY
# Or use keyfile:
luks_keyfile: /path/to/keyfile
luks_mapper_prefix: "hyper2kvm"

# initramfs auto-includes:
#   dm_crypt, xts, dm_mod
```

LVM Handling — Container-Isolated

Two-Phase Activation

- **Phase 1:** Container-isolated VG discovery
- Podman/Docker with private LVM dirs
- No --privileged, targeted capabilities
- --network=none for security
- **Phase 2:** Host-side DM table activation
- Device filter regex per NBD device
- Prevents host VG cache pollution

Safety Features

- Per-job `LVM_SYSTEM_DIR` isolation
- File lock per NBD device (no collisions)
- Stale NBD recovery on startup
- Deactivation of stale VGs before re-activation
- udev settling + dmsetup + 200ms buffer
- 30-second timeout per operation
- Auto-install podman/docker if missing

Storage Support Matrix

LVM2	Full (VG/LV/PV)
mdadm RAID	Software RAID
LUKS2	Auto-unlock
Btrfs	Subvolume-aware
ZFS	Pool import
ext4/XFS	UUID repair
GPT/MBR	Both schemes

Batch Migration & Daemon Mode

Orchestrate hundreds of VMs with parallel workers, checkpoint/resume, and file system watching.

Batch Orchestrator

- Parallel execution with configurable worker limit
- Priority-based VM ordering
- Per-VM error isolation (continue-on-error)
- Shared config + per-VM overrides
- Aggregate progress reporting
- JSON and YAML manifest formats

```
# batch-manifest.yaml
batch_version: 1
batch_metadata:
  batch_id: "vmware-exit-2026"
  parallel_limit: 4
  continue_on_error: true
shared_config:
  output_dir: /data/converted
  out_format: qcow2
  regen_initramfs: true
  emit_domain_xml: true
vms:
  - name: web-01
    vmdk: /exports/web-01.vmdk
    priority: 1
  - name: db-primary
    vmdk: /exports/db-primary.vmdk
    priority: 2
    memory: 16384
    vcpus: 8
```

Checkpoint / Resume

- Automatic checkpoint after each VM completion
- Atomic writes (temp + replace for safety)
- Per-VM granularity — skip completed/failed VMs
- Re-run same command to resume
- Auto-cleanup after successful batch

```
# Checkpoint file (auto-generated):
# .checkpoints/checkpoint-vmware-exit-2026.json
{
  "batch_id": "vmware-exit-2026",
  "completed_vms": ["web-01", "web-02"],
  "failed_vms": ["legacy-app"],
  "resume_from": 3,
  "total_vms": 50
}

# Resume: just re-run the same command
h2kvmctl --batch-manifest batch.yaml
# Skips web-01, web-02, legacy-app
# Continues from VM #4
```

Daemon / Watch Mode

3-Directory Workflow

- **to_be_processed/** — Drop zone for manifests
- **processing/** — Currently processing
- **processed/** — Completed with reports
- **failed/** — Failed with error context
- File system watcher (watchdog library)

Daemon Features

- Concurrent processing (ThreadPoolExecutor)
- Statistics tracking (duration, throughput)
- Detailed reporting with artifact tracking
- Signal handling (SIGTERM, SIGINT)
- Systemd service integration

Systemd Service

```
# hyper2kvm.service
[Unit]
Description=hyper2kvm daemon

[Service]
ExecStart=/usr/local/sbin/h2kvmctl \
  --manifest-workflow-mode \
  --manifest-workflow-dir %i
```

- Supports .json, .yaml, .yml manifests
- File deduplication tracking

- Thread-safe processing queue

Restart=on-failure

```
[Install]
WantedBy=multi-user.target
```

```
# Enable:
systemctl enable --now \
    hyper2kvm@/data/migrations
```

AI Diagnostics & Observability

Machine learning-powered migration intelligence with production-grade structured logging.

AI Migration Intelligence

- 6 sub-engines coordinated by AIOrchestrator
- Fail-safe: exceptions logged, migration continues
- Persistent knowledge base (SQLite)
- Learns from every migration

Diagnostic Engine

- 13 built-in error patterns with regex
- fstab_uuid_mismatch, grub_not_found
- initramfs_missing_drivers, buslogic_no_driver
- vmware_tools_conflict, disk_space_exhausted
- permission_denied, qemu_img_error
- lvm_activation_failed, mount_failed
- Root cause analysis per pattern

Remediation Engine

- Fix registry: pattern_id → fix templates
- Risk levels: LOW, MEDIUM, HIGH
- Auto-applicable flag for automation
- Deduplication and risk filtering
- Suggested commands with explanations

Predictive Engine

- Pre-migration OS/workload prediction
- Success probability scoring
- Risk assessment before starting
- Based on historical migration data

Workload Classifier

- Classifies VM workload type
- Categories: web, database, container, etc.
- Feature extraction for prediction
- Informs fix strategy selection

Anomaly Detector

- Post-migration anomaly detection
- Duration anomaly (too fast/slow)
- Error rate anomaly vs baseline
- Pattern-based identification

Structured Logging & Observability

Canonical Event Schema

```
{
  "ts": "2026-03-24T09:45:14.260+00:00",
  "level": "info",
  "service": "hyper2kvm",
  "component": "disk_processor",
  "host": "kvm-host-01",
  "pid": 12345,
  "trace_id": "50b8fc3c01f9",
  "vm_id": "rhel9-web-01.vmdk",
  "workflow": "disk_processing",
  "span_id": "eb5d54ac",
  "parent_span_id": "a1b2c3d4",
  "event": "flatten_start",
  "phase": "flatten",
  "duration_ms": 45200.5,
```

Production Features

- **Async writes:** Non-blocking with bounded queue + backpressure
- **Rate limiting:** Per-event token-bucket with suppression counts
- **Sampling:** Probabilistic by event-name prefix
- **Field truncation:** Prevents log bloat
- **Safe serialization:** datetime, UUID, Path, bytes
- **Trace propagation:** trace_id, span_id, parent_span_id
- **PhaseTimer:** Context manager for operation timing
- **Audit channel:** Bypasses rate-limit + sampling
- **Dynamic log level:** Runtime control
- **Graceful shutdown:** Queue drain on exit

```
"status": "success"
```

```
}
```

Recovery, Security & Feature Flags

Stage-based recovery with rollback, integrity verification, and gradual feature rollout.

Recovery Manager

- Atomic checkpoint saves (temp + fsync + replace)
- Stage-based recovery (load, inspect, fix, convert)
- Stage order violation detection
- JSON persistence with SHA256 integrity
- POSIX fcntl lock for concurrency
- Crash-safe directory fsync
- Exit codes for automation (20-29)

Snapshot & Rollback

- Snapshot types: FULL, QCOW2, LVM, FILESYSTEM
- Pre-migration snapshot creation
- SHA256 checksum per snapshot
- Snapshot metadata in JSON
- Integrity verification on restore
- Source disk never modified (copy-based)
- Optional `--checksum` for output images

Health Checks

- 8 check types: SYSTEM, SERVICE, NETWORK, DATABASE, APPLICATION, STORAGE, SECURITY, PERFORMANCE
- Status: PASS, FAIL, WARN, SKIP, ERROR
- Remediation suggestions per check
- Post-migration validation (no disk I/O)
- `boot_config`, `fstab_valid`, `virtio_drivers`
- `no_vmware_remnants`, `network_config`

Feature Flags (17 Configurable)

Status	Flag	Description	Control
Stable	vmcraft_backend	Pure Python + qemu-nbd backend (no libguestfs)	Default: ON
Stable	fstab_stabilize	UUID-based fstab rewriting	Default: ON
Stable	initramfs_regen	VirtIO initramfs rebuild	Default: ON
Stable	serial_console	Serial console injection	Default: ON
Stable	buslogic_autofix	BusLogic driver auto-remediation	Default: ON
Beta	ai_diagnostics	AI-powered error diagnosis and remediation	HYPER2KVM_FF_AI_DIAGNOSTICS=1
Beta	multi_drive	Multi-disk VM support	HYPER2KVM_FF_MULTI_DRIVE=1
Beta	parallel_conversion	Parallel qemu-img conversions	HYPER2KVM_FF_PARALLEL_CONVERSION=1
Beta	container_isolation	Container-isolated LVM operations	HYPER2KVM_FF_CONTAINER_ISOLATION=1
Beta	incremental_sync	Incremental disk sync (CBT)	HYPER2KVM_FF_INCREMENTAL_SYNC=1
Exp	luks_auto_unlock	LUKS2 + TPM auto-unlock enrollment	HYPER2KVM_FF_LUKS_AUTO_UNLOCK=1
Exp	tpm_enrollment	TPM2 measured boot enrollment	HYPER2KVM_FF_TPM_ENROLLMENT=1
Exp	network_topology_map	Network topology discovery and mapping	HYPER2KVM_FF_NETWORK_TOPOLOGY_MAP=1

Enterprise-Grade Migration Platform

LUKS2 encryption | Container-isolated LVM | Batch with checkpoint/resume | AI diagnostics
Daemon mode | Structured observability | Recovery + rollback | 17 feature flags

Production-hardened with error handling, timeouts, logging, and concurrency control.